KARTIK SHARMA

Mobile / Android Developer
+91-8929652267 | kartik.dev026@gmail.com
linkedin.com/in/kartik-sharma-1k | kartiksharma1k.github.io

PROFESSIONAL SUMMARY

Mobile / **Android Developer** with **2.5 years** of experience shipping production apps in **Kotlin**, **Java**, **Jetpack Compose** and **Flutter**, with backend experience in **Java** and **Spring Boot**. I build apps that work close to the hardware — camera and **Bluetooth LE** devices, **real-time image processing** and **on-device ML** — and I modernise legacy codebases into clean, modular architectures.

TECHNICAL SKILLS

Kotlin, **Java**, **Dart**, Swift, C++
Android SDK, **Jetpack Compose**, XML Layouts, **Flutter**, SwiftUI
MVVM, MVP, **Clean Architecture**, **Dependency Injection**, **Multi-module Architecture**, Multithreading
REST APIs, **Retrofit**, **gRPC**, Protocol Buffers, **Spring Boot**, Spring Framework, Spring AI
TensorFlow Lite, Core ML, **On-Device ML Inference**
Bluetooth LE, Camera Integration, **Real-Time Image Processing**, **Video Encoding**, Maps SDK, Speech Recognition
Firebase (Firestore, Realtime Database, Authentication), **Room**, SQLite, MongoDB

PROFESSIONAL EXPERIENCE

Software Development Engineer I — Aug 2026 – Present

Nayan India Science and Technology

- Lead **Android** development for a camera-based product, owning app architecture, feature delivery and releases.
- Migrated the production Android codebase to **Kotlin** and from **XML layouts to Jetpack Compose**, adopting state-driven UI and **multithreading** to improve UI performance by over **40%** through fewer frame drops and ANRs.
- Restructured the app into **reusable feature modules**, reducing duplicate code by approximately **35%**.
- Integrated the app with **2 camera devices** (Nayan Cam, KENT) for **real-time image processing** and **video encoding**, cutting processing latency by around **25%**.

Associate Software Engineer — Apr 2024 – May 2026

Hestabit Technologies Pvt. Ltd., Noida, India

- Built an underwater **Android** camera application in **Kotlin**, controlling the camera over **Bluetooth LE**; reduced shutter delay by **40%** through low-latency Bluetooth communication.
- Improved captured image quality by **25%** by implementing on-device **TensorFlow Lite** post-processing, keeping inference on the device instead of a server.
- Developed a voice-enabled golf assistant for **iOS** using **Core ML** and **AVFoundation** voice activity detection, achieving **91.5%** wake-word accuracy and **70%** fewer false activations.
- Refactored **3+ Android and Flutter** applications to **MVVM** with **Dependency Injection**, improving testability and increasing feature delivery speed by **25%**.
- Developed and maintained backend services in **Java** with **Spring Boot** and **Spring Framework**, building and consuming **REST APIs** that powered the mobile applications.
- Integrated **REST APIs** and third-party SDKs including **Maps SDK** and speech recognition, and handled API error states, caching and offline behaviour.
- Designed and published **reusable Kotlin packages** shared across projects, cutting duplicated code by around **40%** and shortening onboarding for new developers.
- Applied **Clean Architecture** across Android and iOS codebases, reducing technical debt by **30%**.

OPEN SOURCE CONTRIBUTIONS

Render Protocol — Jetpack Compose SDK — github.com/renderprotocol

Open-source, AI-powered server-driven UI framework for mobile (MIT License)

- Building the **Android client SDK** with native **Jetpack Compose** renderers that construct UI from **Protocol Buffer** schemas delivered over **gRPC**.
- Contribute to rp-generated-kotlin and rp-utils-kotlin: generated **Kotlin gRPC** stubs, component factories and style resolvers.

numJ — NumPy-inspired Java library — Apache-2.0 | 23 GitHub stars

- Implemented **NDArray** reshape, flatten, transpose and element-wise arithmetic with multi-dimensional broadcasting across **6 numeric data types**.
- Published via **JitPack** for **Gradle** and **Maven**.

PROJECTS

- **Chat Application (Kotlin, Flutter, Firebase):** End-to-end encrypted messaging with disappearing messages and phone authentication; rated **4.8 stars** across **500+ Play Store reviews**.
- **Video Player (Android):** Custom player with picture-in-picture, ad skipping and **Firebase** caching for faster data loading.
- **E-Kites Exam Prep (Jetpack Compose):** JEE/NEET preparation app with personalised tests, timers, performance analytics and a custom math-rendering UI.

EDUCATION

B.Tech, Information Technology — **2021 – 2024**
Dr. Akhilesh Das Gupta Institute of Technology & Management, Delhi — *CGPA: 8.41 / 10*

Diploma, Computer Science — **2018 – 2021**
Ambedkar Institute of Technology, Delhi — *CGPA: 7.4 / 10*

ACHIEVEMENTS

- **Top 5 team** in a government-organised on-campus hackathon.
- Ranked **3rd out of 60** students in the IT branch in the campus placement drive.